

AI Career Coach: A Multi-Agent Conversational System for Resume Optimization and Interview Preparation

Abstract

This report presents the design and implementation of *AI Career Coach*, a command-line intelligent assistant built using the Moya multi-agent framework and OpenAI's GPT models. The system supports resume analysis, ATS scoring, optimization, and mock interview simulations. By leveraging agent specialization, contextual memory, and a tool-based execution framework, the assistant provides personalized and structured feedback to users preparing for job applications. The system combines document processing, conversational memory, classification, and orchestration to create a holistic career preparation experience.

Introduction

Job seekers often struggle to tailor their resumes and prepare effectively for interviews. Traditional tools lack personalization, memory of past interactions, and conversational support. The AI Career Coach bridges this gap by integrating resume feedback and mock interviews into a multi-agent dialogue system powered by LLMs.

Using OpenAI's `gpt-4o` model and the Moya framework, the system facilitates natural interactions with specialized agents while maintaining conversation context. This provides job seekers with continuous guidance, actionable suggestions, and mock interview feedback—all from a single interface.

Methodology

1. Document Parsing

- **Input Format:** DOCX and PDF files
- **Libraries:** `docx` for DOCX and `PyPDF2` for PDFs

- **Purpose:** Extract plaintext from resumes and job descriptions

2. Memory Tool

- **Custom Unified Tool:** Implements `store_message`, `get_last_n_messages`, and `get_thread_summary`
- **Backend:** Uses `InMemoryRepository` for temporary thread-based message storage
- **Usage:** Agents retrieve prior user interactions to maintain context and continuity

3. Agent Definitions

- **Resume Agent:** Analyzes resumes, provides ATS score, and recommends improvements
- **Interview Trainer Agent:** Conducts mock interviews with structured feedback after each answer
- **Classifier Agent:** Routes user inputs to the appropriate agent based on intent (resume vs. interview)

4. Tool Registry

- A centralized registry registers the unified memory tool and makes it accessible to all agents

5. Multi-Agent Orchestration

- **Framework:** `MultiAgentOrchestrator` from Moya
- **Classifier:** LLM-based routing using structured prompts
- **Default Agent:** `resume_agent`

Inference Flow

1. **User Input:** Command-line prompt `/resume path` or natural query
2. **Classifier Agent:** Chooses between `resume_agent` or `interview_trainer_agent`
3. **Memory Tool:**
 - Stores the user message
 - Retrieves the last 5 messages and a short summary
4. **Prompt Generation:** Constructs prompt with resume, JD, conversation summary, and user input
5. **Agent Inference:** Generates structured response using `gpt-4o`
6. **Streaming Output:** Agent response is streamed and printed
7. **Repeat:** Loop continues until `exit` is typed

Features Implemented:

- Resume and JD ingestion (`.docx`, `.pdf`)
- Unified memory tool with store/retrieve/summary functionality
- ATS scoring and actionable feedback generation
- Mock interview with 5–10 questions, live feedback, and scoring
- Context-aware conversation using stored history
- Agent routing based on user query (classifier agent)
- Real-time streaming chat interface in CLI

Results:

GREETING MESSAGE FROM THE AGENT

👋 Welcome to Your AI Career Coach!

Ready to land your dream job?

We're here to build your perfect resume and help you ace interviews with confidence.



Tailored Resume Builder

Just tell us about your experience and the job you're aiming for—we'll craft a polished, customized resume in minutes.



Mock Interview Trainer

Practice with realistic interview questions, get real-time feedback, and track your progress with smart memory insights.

Let's get started—your career breakthrough begins here!

Upload resume and job description as:

/resume path/to/resume.docx or .pdf

/jd path/to/job_description.docx or .pdf

Type 'exit' to quit.

You: /resume

/kaggle/input/sample-resumes/Rajesh_Kumar_Weak_ATS_Resume.docx



Resume uploaded.

You: /jd /kaggle/input/sample-jds/Data Analyst Entry Level.pdf



Job description uploaded.

ATS SCORE AND SUGGESTIONS FOR THE RESUME

You: can you review my resume and give ats score?

Assistant: [resume_agent] ### ATS Score Analysis:

ATS Score: **68/100**

While your resume shows potential and mentions critical skills and educational qualifications, it does not adequately align sales trends".

4. Include Another Section for Tools/Technologies:

To align with the job description, explicitly mention tools/software in a "Technical Tools" section:

Examples:

- **Proficient in**: MS Excel, SQL, Python
- **Exposure to**: Power BI, Tableau, Data Cleaning Techniques

5. Rephrase Objective:

Replace the vague objective with a tailored statement for the Data Analyst role:

Example:

"To apply analytical and problem-solving skills to provide actionable data-driven insights as a Data Analyst. Skilled in Python, SQL, and Excel with a keen interest in leveraging data visualization tools to solve business challenges."

Modified Bullet Point Suggestions:

Experience Section:

Revise from:

- Helped with system maintenance.
- Assisted in lab inventory.
- Supported students during lab hours.

To something like:

- Supported system maintenance by troubleshooting technical issues and ensuring lab systems were operational 95% of the time.
- Organized and managed lab inventory, ensuring a 10% reduction in missing or misplaced assets through effective tracking methods.
- Assisted 50+ students in data handling and IT-related tasks, fostering learning through hands-on guidance.

Projects Section:

Revise from:

1. Data Cleaning

- Cleaned a dataset in Excel.
- Removed missing values and duplicate rows.

To something like:

1. **Data Cleaning Project**

- Cleaned and preprocessed a dataset with 10,000+ records using **MS Excel**, handling missing values and reducing redundancy by 15%.
- Generated preliminary insights by analyzing trends in the cleaned data.

****Skills Section:****

Expand from:

- MS Excel, Python, SQL, Team work, Communication, Computer basics

To something like:

- ****Programming & Analysis****: Python, SQL
- ****Visualization Tools****: Exposure to Power BI, Tableau
- ****Data Management****: MS Excel (Advanced), Data Cleaning, Dataset Handling

- ****Soft Skills****: Team Collaboration, Communication, Analytical Thinking

Focus on Quantifiable Impacts:

- Include metrics for projects (number of rows, time saved, efficiency improvement, etc.).
- Use ****action verbs**** such as "analyzed," "built," "optimized," "streamlined."

Additional Suggestions:

1. Add a ****"Professional Summary"**** right after the name/contact details.

Example: "Entry-level Data Analyst with a B.Sc. in Mathematics and hands-on experience in data cleaning, preprocessing, and visualization using Python, SQL, and Excel. Skilled in interpreting trends and driving actionable insights."

2. Add a ****Technical Skills**** section distinct from "Soft Skills."

3. Certifications such as "Python for Beginners" could be more specific. Mention the improved expertise or projects you completed as part of the certification.

Next Steps:

Include these updates and realign the resume to the job description for potentially increasing the ATS score. Once you've updated the resume, feel free to share it again for further improvements.

OPTIMISED RESUME AND ATS SCORE AFTER OPTIMISATION

You: can you give optimised resume and the score after optimisation

Assistant: [resume_agent] ### Initial ATS Analysis:
Your resume seems broadly aligned with fields related to data analytics.
However, some key points from the job description aren't adequately
reflected in the resume. Here is the breakdown:

ATS Score: **65/100**

Key Gaps Noticed:

1. **Missing Keywords and Tools**: Examples include "data visualization tools" (Tableau, Power BI), "problem-solving skills," and "dashboards."
2. **Project Details Lack Depth**: Projects do not explain measurable results or methods/tools used (e.g., Python, SQL, data visualization software).
3. **Lack of Metrics and Impact**: Achievements and skills are listed without context, numbers, or demonstrable outcomes.
4. **Job-Specific Terminology Missing**: Skills mentioned like "trends," "anomalies," and "cleaning data with SQL" from the job description are absent.
5. **Objective Statement**: Does not explicitly express enthusiasm for data analytics or solving data problems.

Optimized Resume:

Rajesh Kumar

Email: rajesh123@gmail.com | **Phone**: +91-9876543210 | **LinkedIn**:
linkedin.com/in/rajeshkumar

Objective

Passionate and detail-oriented **Data Analyst** eager to utilize skills in **Python**, **SQL**, and **data visualization tools** to derive actionable insights. Committed to solving complex analytical challenges and contributing to business growth at a reputed organization like Fractal Analytics.

Education

B.Sc. in Mathematics

XYZ College, Delhi University | **Graduated**: 2023 | **CGPA**: 7.2

Technical Skills

- **Programming**: Python, SQL
- **Data Visualization**: MS Excel, Tableau (familiar)
- **Soft Skills**: Effective communication, Team collaboration, Problem-solving
- **Other**: Data Cleaning and Transformation

Experience

****Intern | Computer Lab Assistant****

ABC Institute (Jan 2023 - Apr 2023)

- Supported system maintenance and ensured smooth operations, leading to increased lab efficiency.
- Assisted with lab inventory tasks, improving inventory management by 15%.
- Guided students with troubleshooting issues during system usage.

Projects

****1. Data Cleaning and Analysis in E-commerce Data****

- Cleaned a raw dataset with over 10,000 entries, using ****Python pandas**** and Excel to handle missing data and inconsistencies.
- Built interactive sales performance dashboards in Excel, showcasing trends and anomalies.

****2. Regional Sales Insights Dashboard****

- Analyzed regional sales trends in mock datasets using ****SQL**** for data manipulation and aggregation.
- Leveraged Tableau (trial) to create dynamic dashboards presenting key sales KPIs, improving decision-making efficiency.

Certifications

- ****Python for Everybody Specialization**** - Coursera
- ****Excel Basics**** - Udemy

Achievements

- Finalist in College Data Visualization Contest, showcasing ****Excel-based dashboards****.
- Organized a predictive analytics game for the college annual fest, enhancing peers' understanding of data applications.

Languages

- English (fluent)
- Hindi (native)

ATS Score After Optimization: **92/100**

Key Improvements Made:

1. **Added Relevant Keywords**: Incorporated "Tableau," "dashboards," "data visualization," "Python pandas," "problem-solving," and "SQL aggregation" to align with the job description.
2. **Expanded on Projects with Tools**: Clarified tools and techniques used, such as Python, Excel, SQL, and Tableau with quantitative impacts.
3. **Described Skills and Achievements in Managerial Language**: Added measurable outcomes (e.g., increasing inventory management efficiency, cleaning datasets).
4. **Tailored Objective Statement**: Made it more focused on the data analyst role and the job description.

INTERVIEW PREPARATION AGENT SWITCH FROM RESUME BUILDER AGENT

You: i want to practice for interview for this role

Assistant: [interview_trainer_agent] Hello Rajesh! Great to see that you're preparing for the Data Analyst position at Fractal Analytics—it's an excellent opportunity for fresh graduates like you!

Before we begin, could you please let me know what type of interview questions you'd like to focus on today? For example:

1. HR Questions (e.g., "Tell me about yourself")
2. Behavioral Questions (e.g., "Describe a challenge you faced and how you overcame it")
3. Technical Questions (e.g., Python, Excel, SQL, data analysis tools)
4. Mixed Questions (a combination of the above)

Let me know, and we can get started. Are you ready?

You: technical questions

Assistant: [interview_trainer_agent] Hello Rajesh! Let's begin preparing for technical questions for the Data Analyst - Entry Level role at Fractal Analytics. Are you ready to dive in? I'll ask one question at a time and provide detailed feedback after each of your answers. Let me know when you're ready!

You: yeah i am ready

Assistant: [interview_trainer_agent] Great, Rajesh! Let's kick off this mock interview focusing on technical questions tailored to the Entry-Level Data Analyst role at Fractal Analytics. After each question, I'll provide feedback to help you refine your responses.

Here's the first question:

Question 1:

Can you explain the differences between primary keys and foreign keys in a database?

Feel free to take your time and respond thoughtfully!

You: "Primary keys and foreign keys are the same thing. They are both used to make sure that tables in a database are connected. A primary key is any column in a table, and a foreign key is just another term for the same thing but in a different table. You can have as many primary keys as you want in one table, and foreign keys don't really need to match anything. They just help store extra information."

FEEDBACK FROM THE INTERVIEWER AGENT

Assistant: [interview_trainer_agent] Alright, let's evaluate your answer, starting with feedback:

Structured Feedback

****Strength:****

- You have correctly identified that primary keys and foreign keys are related to connecting tables in a database.

****Weakness:****

- While primary keys and foreign keys are connected concepts, they are not the same thing.
- A primary key uniquely identifies rows in a table, while a foreign key is used to establish and enforce a link between the data in two tables.
- The statement that "you can have as many primary keys as you want in a table" is not correct. Each table can only have one primary key, though that key can sometimes comprise multiple columns (composite key).
- Foreign keys must match the primary key of the other table if the relationship is to maintain referential integrity.

****Improvement Tip:****

A better explanation would clarify the difference between primary keys and foreign keys:

- A ****primary key**** is a unique identifier for each row in a table (e.g., a "Student ID" column).
- A ****foreign key**** is a field (or set of fields) in one table that refers to the primary key in another. For example, if "Student ID" from a `Students` table is referenced in a `Grades` table to link grades to a specific student, the "Student ID" in `Grades` would be a foreign key.

Next question: *****Can you explain what a JOIN is in SQL? Provide a brief example.*****

KEEPING TRACK OF LAST MESSAGES AND CONVERSATIONS

--- Last 5 messages ---

```
[{'role': 'user', 'content': 'can you give optimised resume and the score after optimisation', 'timestamp': '2025-06-18T10:45:28.447406', 'metadata': {}}, {'role': 'user', 'content': 'i want to practice for interview for this role', 'timestamp': '2025-06-18T10:47:28.886075', 'metadata': {}}, {'role': 'user', 'content': 'technical questions', 'timestamp': '2025-06-18T10:47:57.864457', 'metadata': {}}, {'role': 'user', 'content': 'yeah i am ready', 'timestamp': '2025-06-18T10:48:22.829602', 'metadata': {}}, {'role': 'user', 'content': '"Primary keys and foreign keys are the same thing. They are both used to make sure that tables in a database are connected. A primary key is any column in a table, and a foreign key is just another term for the same thing but in a different table. You can have as many primary keys as you
```

```
want in one table, and foreign keys don\'t really need to match anything.
They just help store extra information."', 'timestamp':
'2025-06-18T10:49:09.444408', 'metadata': {}}}
```

--- Conversation Summary ---

user: can you review my resume and give ats score?

user: can you give optimised resume and the score after optimisation

user: i want to practice for interview for this role

user: technical questions

user: yeah i am ready

user: "Primary keys and foreign keys are the same thing. They are both used to make sure that tables in a database are connected. A primary key is any column in a table, and a foreign key is just another term for the same thing but in a different table. You can have as many primary keys as you want in one table, and foreign keys don't really need to match anything. They just help store extra information."

Conclusion

The AI Career Coach demonstrates the potential of multi-agent conversational systems in delivering structured career coaching. By integrating resume analytics, mock interviews, and contextual memory into a unified assistant, it offers a significant enhancement over static resume builders or interview prep tools. With modular architecture, the system is extensible for additional agents (e.g., job search agents, salary advisors), advanced memory backends (vector stores), and graphical user interfaces (e.g., Streamlit). This architecture lays a foundation for scalable, personalized career development experiences powered by language models.